

PillSync

1. Title: PillSync: Intelligent Medicine Reminder and Medication Tracking Platform

2. Objective

Build an intelligent medicine reminder and medication tracking platform that helps users manage medicine schedules, monitor dosage adherence, predict refill requirements, and maintain medication history using AI-powered tracking and smart notifications.

The system should allow patients and caregivers to manage medications, schedule reminders, track medicine consumption, receive refill alerts, and maintain long-term treatment records.

The platform is designed to support chronic disease management by improving medication adherence, reducing missed doses, and ensuring timely medicine refills through intelligent automation.

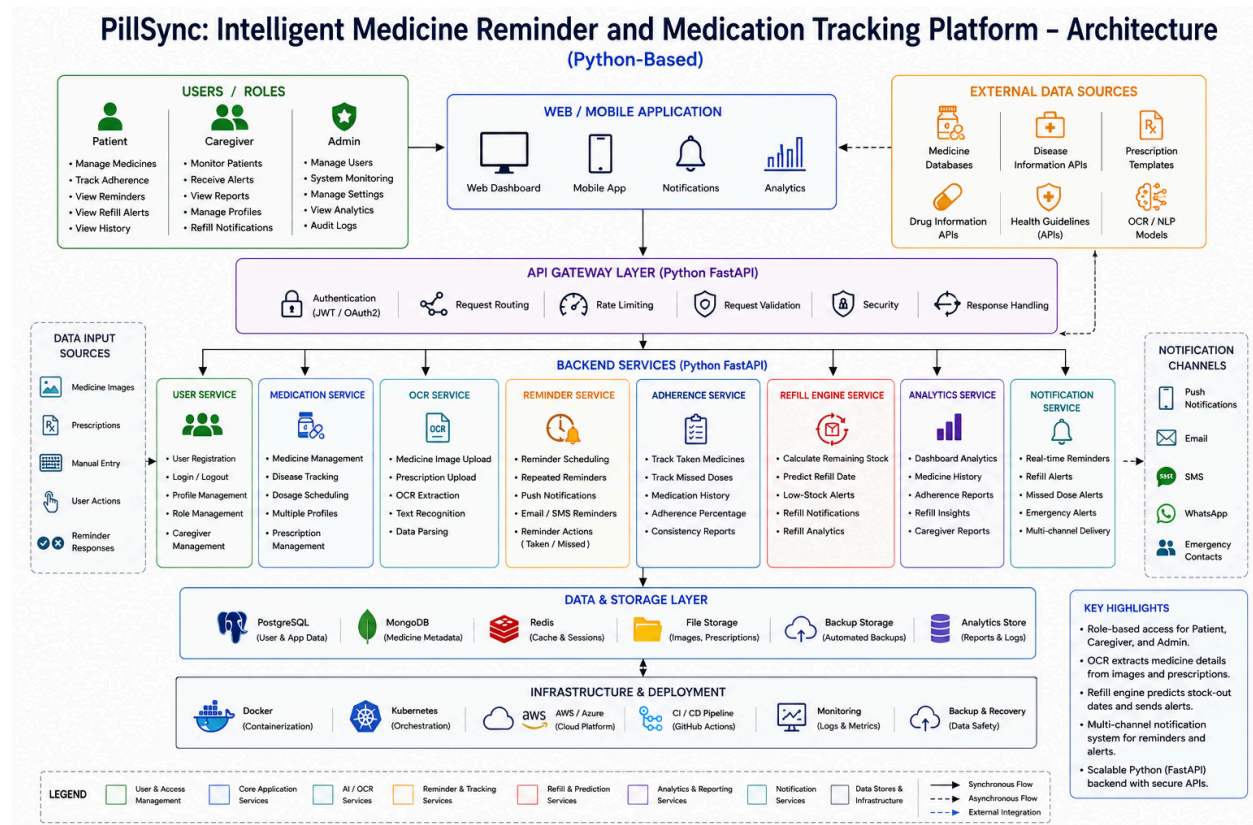
This solution can be used for personal healthcare management, family medication tracking, caregiver monitoring, elderly care support, and chronic disease treatment management.

Outcomes

- Developed and deployed an intelligent medication management platform.
- Implemented secure authentication and role-based access control.
- Built medicine upload and medication scheduling workflows.
- Integrated OCR-based medicine recognition and prescription extraction.
- Developed medication adherence tracking and reporting systems.
- Implemented AI-powered refill prediction and low-stock alert mechanisms.
- Built dashboards for medication history and adherence analytics.
- Implemented caregiver monitoring and notification workflows.

- Deployed the platform using Docker and cloud deployment platforms such as AWS or Azure.

3. Architecture Diagram:



4. Modules to be Implemented

1. User Authentication & Role-Based Access

Authentication Features

- JWT Authentication
- OAuth2 Login
- Session Management
- Secure Password Management

Role Management

Patient

- Manage personal medicine schedules
- Upload prescriptions and medicine details
- Track medication adherence
- View refill predictions
- Receive medication reminders
- View medication history

Caregiver

- Monitor assigned patients
- Receive missed-dose alerts
- Receive refill notifications
- View adherence reports
- Manage multiple patient profiles

Admin

- Manage users and caregiver assignments
- Monitor platform activities
- Configure notification settings
- Access platform analytics
- Manage system-level operations

2. User Profiles & Medication Management

- Profile creation and management
- Medicine management
- Disease and condition tracking

- Dosage scheduling
- Prescription management
- Multiple patient profiles for families

3. Medicine Upload & OCR Recognition

Users can:

- Upload medicine images
- Upload prescriptions
- Manually enter medicine details

OCR extracts:

- Medicine name
- Dosage
- Quantity
- Frequency
- Prescription details

4. Smart Reminder System

- Morning reminders
- Afternoon reminders
- Night reminders
- Repeated reminder scheduling
- Push notifications
- Email reminders
- SMS reminders

Reminder Actions:

- Medicine Taken
- Medicine Missed

- Snooze Reminder

5. Medication Adherence Tracking

- Track taken medicines
- Track missed doses
- Daily medication history
- Adherence percentage calculation

Dashboard Analytics

- Consistency tracking
- Weekly medication reports
- Monthly medication reports
- Missed dosage analysis
- Adherence trends

6. AI Refill Prediction Engine

Refill Engine Workflow

The refill engine continuously monitors medication stock levels using dosage schedules and medicine consumption records.

Input Parameters

- Initial medicine quantity
- Daily dosage frequency
- Quantity consumed per dose
- Missed dosage history
- Manual stock updates

Refill Prediction Logic

The system calculates:

- Remaining stock
- Average daily consumption
- Estimated stock depletion date
- Recommended refill date

Refill Features

- Automatic stock calculation
- Refill date prediction
- Low-stock alerts
- Refill reminders
- Caregiver refill notifications
- Refill analytics dashboard

Example

Medicine Quantity: 60 Tablets

Dosage: 2 Tablets Per Day

Estimated Refill Requirement:

$$60 \div 2 = 30 \text{ Days}$$

Notification Example:

"Your BP medicine is expected to finish in 5 days. Please arrange a refill."

7. Disease-Based Medication Organization

Organize medicines by:

- Blood Pressure
- Diabetes
- Thyroid
- Antibiotics
- Vitamins

- Heart Medications

8. Smart Notifications & Alerts

- Real-time medicine reminders
- Refill alerts
- Missed medicine notifications
- Emergency caregiver notifications
- Prescription expiry reminders

9. Dashboard & Analytics

- Medicine history
- Refill tracking
- Adherence analytics
- Active medicine list
- Health consistency reports
- Caregiver monitoring reports
- Refill prediction analytics

10. Final Integration, Testing & Deployment

- Module integration
 - End-to-end testing
 - Performance validation
 - Production deployment
 - Documentation and demonstration
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5. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Requirements, Database Design & Core Setup

- Define project scope and healthcare workflows.
- Design database schema.
- Create UI wireframes and workflow planning.
- Setup frontend and backend environments.
- Implement authentication and role-based access control.
- Configure PostgreSQL database.
- Build user profile management features.

Outcomes

- Understand healthcare workflow management concepts.
- Learn database design and system architecture.
- Build frontend and backend project initialization.
- Working authentication and profile management system.

Milestone 2: Week 3 & 4 — Medication Management & Reminder System

- Develop medicine management workflows.
- Build dosage scheduling features.
- Implement medicine reminder system.
- Integrate push notifications.
- Build medication history tracking.

Outcomes

- Implement medicine scheduling and tracking workflows.
- Build reminder and notification systems.
- Understand medication adherence concepts.

- Generate medication history reports.

Milestone 3: Week 5 & 6 — OCR Recognition & Refill Prediction

- Implement OCR-based medicine recognition.
- Build refill prediction engine.
- Develop dosage analysis workflows.
- Generate refill notifications.
- Implement adherence analytics.

Outcomes

- OCR medicine recognition operational.
- Refill prediction system functional.
- Medication adherence tracking completed.
- Intelligent refill management workflows implemented.

Milestone 4: Week 7 & 8 — Analytics, Testing & Deployment

- Build analytics dashboard.
- Add refill and adherence visualizations.
- Perform testing and validation.
- Deploy platform using cloud environments.
- Prepare final documentation and presentation.

Outcomes

- Fully deployed production-ready application.
 - Reminder and refill systems operational.
 - Analytics dashboard functional.
 - Complete end-to-end medication management workflow.
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6. Evaluation Criteria

Milestone 1 (Week 2)

- Backend initialization completed.
- Authentication workflows implemented.
- Database schema finalized.
- Frontend setup completed.

Milestone 2 (Week 4)

- Medicine management operational.
- Reminder scheduling system functional.
- Medication history tracking implemented.
- Notification workflows integrated.

Milestone 3 (Week 6)

- OCR medicine recognition operational.
- AI refill prediction system functional.
- Medication adherence tracking completed.
- Refill notifications working correctly.

Milestone 4 (Week 8)

- Fully deployed frontend and backend.
 - Analytics dashboards operational.
 - Testing and validation completed.
 - End-to-end medication workflow demonstrated.
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7. Tools & Tech Stack

Programming Language

Backend: Python (Django REST Framework / FastAPI)

Frontend: React.js

Database

- PostgreSQL
- SQLite (Development)

AI & OCR Technologies

- Tesseract OCR
- spaCy
- OpenAI API

Authentication

- JWT Authentication
- OAuth2 Authentication

Cloud & DevOps

- Docker
- AWS / Azure
- GitHub Actions

Libraries & Frameworks

- Django
- Django REST Framework
- FastAPI
- React.js
- Tailwind CSS
- Axios
- Redux / Context API

- Firebase Cloud Messaging

Notification Services

- Firebase Cloud Messaging
- Twilio
- SendGrid

Testing Tools

- Pytest
- Django Test Client
- Jest
- React Testing Library

Dev & Deployment Tools

- VS Code
 - Git & GitHub
 - Docker & Docker Compose
 - Postman
 - Render / AWS / Vercel
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8. Performance Metrics

Medication Tracking Performance

- Medication adherence accuracy
- Reminder delivery success rate
- Missed-dose detection accuracy

Refill Prediction Performance

- Refill prediction accuracy

- Low-stock alert accuracy
- Refill notification effectiveness

Analytics Performance

- Dashboard response time
- Report generation performance
- Data processing efficiency

System Performance

- API response time
 - Concurrent user handling
 - Database query optimization
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9. Example Quantitative Goals

Medication Tracking

Achieve accurate medication adherence monitoring and reminder tracking.

Refill Prediction

Generate reliable refill predictions and low-stock alerts before medicine depletion.

Analytics

Provide actionable medication adherence and refill insights.

Platform Performance

Support multiple users and caregivers with stable medication management workflows.